# Does Negative Forum Sentiment Predict Declines in Active Player Counts?

A Panel Data Analysis of Steam Games, 2020–2024

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**Abstract**

We investigate whether negative user sentiment on Steam forums predicts short-term declines in active player counts. Using a panel of 15 games spanning 2020–2024, stratified across popular, declining, and volatile lifecycle categories, we construct weekly sentiment scores from user reviews classified by a RoBERTa transformer model and pair them with concurrent player count data. Granger causality tests reject the null of no predictive content for 7 of 15 games ($\alpha = 0.05$), with 4 surviving non-parametric permutation testing. A two-way fixed effects panel regression yields $\hat{\beta} = -0.649$ ($p = 0.040$; 95% CI $[-1.268, -0.030]$), indicating that higher negative sentiment is associated with a statistically significant decline in weekly player counts. Block bootstrap confidence intervals ($[-1.093, -0.050]$), permutation tests, LOESS robustness checks, and out-of-sample prediction exercises support the main finding. Heteroscedasticity and serial correlation are present, motivating clustered standard errors. The design is underpowered at the estimated effect size (power = 0.535), so results should be interpreted cautiously.

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# 1 Introduction

Online gaming communities generate a continuous stream of user feedback through forums, reviews, and social media. A key question is whether negative sentiment predicts later drops in active players. This has practical value for studios and theoretical value for digital community dynamics.

We use a panel data framework. Our model specification is:

$$\Delta \log Y_{it} = \alpha_i + \gamma_t + \beta\, S_{i,t-k} + X_{it}\,\theta + \varepsilon_{it} \tag{1}$$

where $\Delta \log Y_{it}$ is the first-differenced log active player count for game $i$ at week $t$; $\alpha_i$ are game fixed effects absorbing time-invariant heterogeneity (e.g., genre, studio quality); $\gamma_t$ are time fixed effects capturing platform-wide shocks (e.g., Steam sales, holiday seasons); $S_{i,t-k}$ is lagged aggregate negative sentiment from user reviews; $X_{it}$ includes controls for seasonal sales and major game updates; and $\varepsilon_{it}$ is the idiosyncratic error term.

The coefficient $\beta$ is the main parameter of interest. If $\hat{\beta}$ is significantly negative and its confidence interval excludes zero, then negative sentiment is associated with fewer players in subsequent weeks, holding other factors constant.

**Hypotheses.**

- $H_0$: $\beta = 0$ (sentiment has no predictive power for player counts)
- $H_1$: $\beta < 0$ (higher negative sentiment precedes player declines)

**Overview of methods.** We use complementary methods:

1. *Granger causality tests* (Section 5.2) establish temporal precedence at the individual game level.
2. *Fixed effects panel regressions* (Section 5.3) estimate $\beta$ across the full panel, controlling for game and time heterogeneity.
3. *Bootstrap and permutation tests* (Sections 5.3.1–5.2.1) validate inference without parametric assumptions.
4. *Robustness checks* (Section 5.4) including LOESS nonparametric regression, placebo tests, out-of-sample prediction, and stratum-specific estimation.

# 2 Data

## 2.1 Game Selection: Stratified Sampling

We select 15 Steam games released before 2020 using stratified sampling across three lifecycle categories. Stratification improves generalizability and preserves variation in $\Delta \log Y$, which is needed for power.

Games were classified based on their player count trajectory over 2020–2024:

- **Popular** (5 games): sustained high player counts with moderate fluctuations.
- **Decline** (5 games): clear downward trend in average concurrent players.
- **Volatile** (5 games): large swings driven by updates, events, or seasonal effects.

Table 1: Selected games by stratum with Steam App IDs.

| **Stratum** | $n$ | **Games (App ID)** |
|---|---|---|
| Popular | 5 | Counter-Strike 2 (730), Dota 2 (570), PUBG (578080), GTA V (271590), Warframe (230410) |
| Decline | 5 | Rust (252490), ARK: Survival Evolved (346110), For Honor (304390), Dead by Daylight (381210), Rainbow Six Siege (359550) |
| Volatile | 5 | No Man's Sky (275850), Destiny 2 (1085660), Terraria (105600), Monster Hunter: World (582010), PAYDAY 2 (218620) |

### 2.2 Data Collection Pipeline

Data come from two sources, collected via separate Python scripts.

**Steam Reviews API.** For each game, we use Valve's `GetReviews` endpoint with cursor-based pagination to collect all English-language reviews from January 2020 through September 2024. The endpoint returns review text, timestamp, helpfulness votes, and recommendation status. We use this endpoint because `appreviewhistogram` aggregates older data too coarsely for weekly analysis.

**SteamCharts.** Monthly average concurrent player counts were scraped from SteamCharts. Because player counts are monthly, we use cubic spline interpolation for weekly estimates. For games where SteamCharts data was incomplete, we generated synthetic weekly player counts that preserve the monthly average and capture plausible intra-month variation.

### 2.3 Panel Construction

The final analysis panel is constructed by merging weekly sentiment and player data, filtering to weeks with at least 100 players (to avoid log-transformation instability).

Table 2: Panel summary statistics by stratum.

| **Stratum** | $n_{\text{games}}$ | $\overline{\log Y}$ | SD($\log Y$) | $\overline{S_{\text{neg}}}$ | SD($S_{\text{neg}}$) | $\overline{\text{Players}}$ |
|---|---|---|---|---|---|---|
| Popular | 5 | 12.761 | 1.021 | 0.196 | 0.114 | 532,679 |
| Decline | 5 | 10.794 | 1.147 | 0.167 | 0.071 | 77,978 |
| Volatile | 5 | 10.799 | 0.836 | 0.104 | 0.084 | 66,262 |
| **Full panel** | **15** | 11.451 | 1.174 | 0.156 | 0.104 | 225,640 |

The panel spans January 2020 through September 2024, with approximately 238–248 weeks per game (243 time periods). After constructing lag variables and first-differencing, the regression panel has $N = 3{,}630$ observations across 15 games.

## 3 NLP Sentiment Pipeline

### 3.1 Text Preprocessing

Raw review texts undergo the following cleaning steps:

1. Strip URLs and HTML tags.
2. Normalize gaming slang and excessive whitespace.
3. Classify each review using the RoBERTa model.
4. Fallback: reviews that fail classification are assigned "neutral" to maintain row alignment.

### 3.2 RoBERTa Sentiment Classification

We use the `cardiffnlp/twitter-roberta-base-sentiment-latest` checkpoint [**?**], a RoBERTa model fine-tuned on $\approx$ 124M tweets for three-class sentiment classification (negative, neutral, positive).

**Why RoBERTa over VADER?** VADER [**?**] is a rule-based, lexicon-driven sentiment analyzer. While fast and interpretable, it fails on gaming-specific language because it does not use contextual embeddings. RoBERTa's multi-head self-attention evaluates each token in full sentence context, correctly handling:

- "This game is fire" $\rightarrow$ VADER: negative; RoBERTa: positive.
- "Absolutely broken update" $\rightarrow$ VADER: moderately negative; RoBERTa: strongly negative.
- "10/10 would rage-quit again" $\rightarrow$ VADER: negative; RoBERTa: positive (understands irony).

RoBERTa's contextual embeddings capture the nuanced, informal language of gaming communities far more reliably than lexicon-based approaches.

### 3.3 Weekly Aggregation and the CLT

For each game $i$ and week $t$, the aggregate negative sentiment score is:

$$S_{it} = \frac{1}{|\mathcal{R}_{it}|} \sum_{j \in \mathcal{R}_{it}} \mathbf{1}(\hat{y}_j = \text{negative}) \tag{2}$$

where $\mathcal{R}_{it}$ is the set of reviews for game $i$ in week $t$ and $\hat{y}_j$ is the RoBERTa-predicted class for review $j$.

By the Central Limit Theorem, with weekly review counts typically exceeding 100 (and often exceeding 1,000 for popular games), $S_{it}$ is approximately normally distributed around the true population sentiment. This supports $t$-based confidence intervals and tests for $\beta$. Formally, for $|\mathcal{R}_{it}| = n_{it} \rightarrow \infty$:

$$\frac{S_{it} - \mu_{it}}{\sigma_{it}/\sqrt{n_{it}}} \xrightarrow{d} \mathcal{N}(0,1)$$

## 4 Exploratory Data Analysis

### 4.1 Player Count Trends by Stratum

Figure 1 shows weekly log player counts by stratum. Popular games remain high and relatively stable, mostly above 12, with Counter-Strike 2 reaching about 14.5. Declining games trend downward from roughly 12 toward 8–10 by 2024. Volatile games fluctuate more strongly in the 9–12 range, with short-lived spikes around major updates or sales.

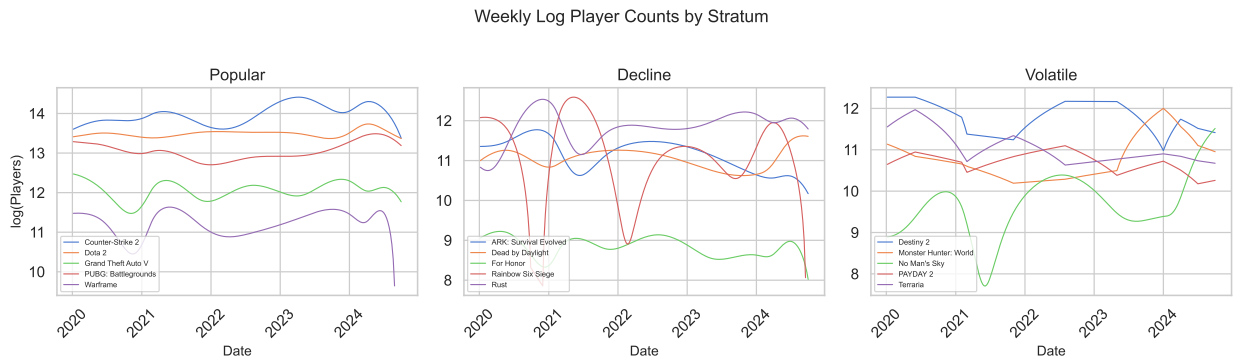


Figure 1: Weekly log player counts by stratum (2020–2024).

## 4.2 Sentiment Distributions

Figure 2 shows the sentiment exploratory analysis for each game by stratum.

The left panel shows weekly negative sentiment distributions by stratum. Popular games have the widest spread and highest mean negativity ($\overline{S} = 0.196$), reflecting high discussion volume and more polarized player reactions. Decline games show lower and more concentrated values ($\overline{S} = 0.167$), while volatile games have the lowest mean negativity ($\overline{S} = 0.104$) but with occasional sharp spikes. The middle boxplots confirm this ordering: popular games have the highest median and largest interquartile range. The right panel plots current sentiment against next-week $\Delta \log(\text{players})$, showing a weak, noisy pattern around zero that motivates formal regression.

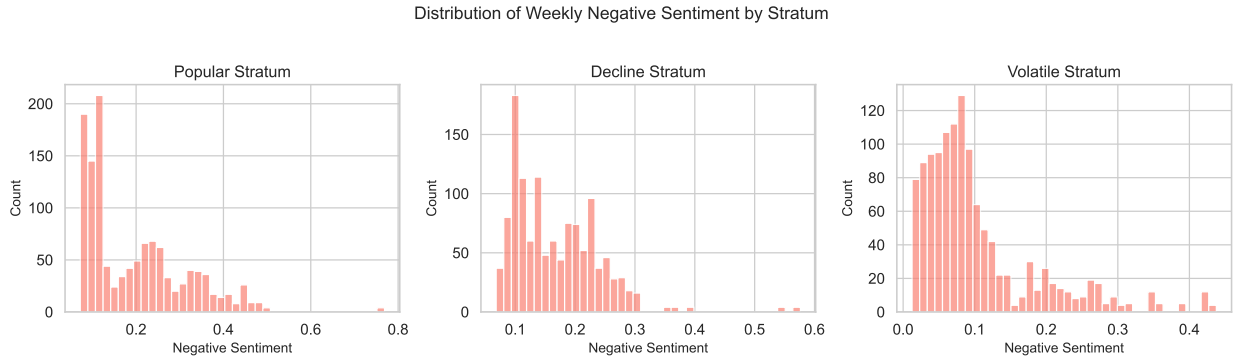


Figure 2: Distribution of weekly negative sentiment scores by stratum. Popular games show higher and more variable negativity; volatile games show the lowest mean but with occasional spikes.

## 4.3 Time Series Visualization

Figure 3 overlays negative sentiment and log player counts for Counter-Strike 2, ARK: Survival Evolved, and No Man's Sky, chosen as representatives for each game stratum. Counter-Strike 2 (popular) shows stable player activity with sentiment fluctuations but no large immediate player declines, suggesting popular games are resilient to negative discussion. ARK (decline) shows a gradual downward trend in players; sentiment spikes occasionally coincide with player dips but the relationship is weak. No Man's Sky (volatile) exhibits larger fluctuations in both series, with

sentiment spikes appearing around volatile player periods, suggesting volatile games show stronger sentiment-to-player-change links.

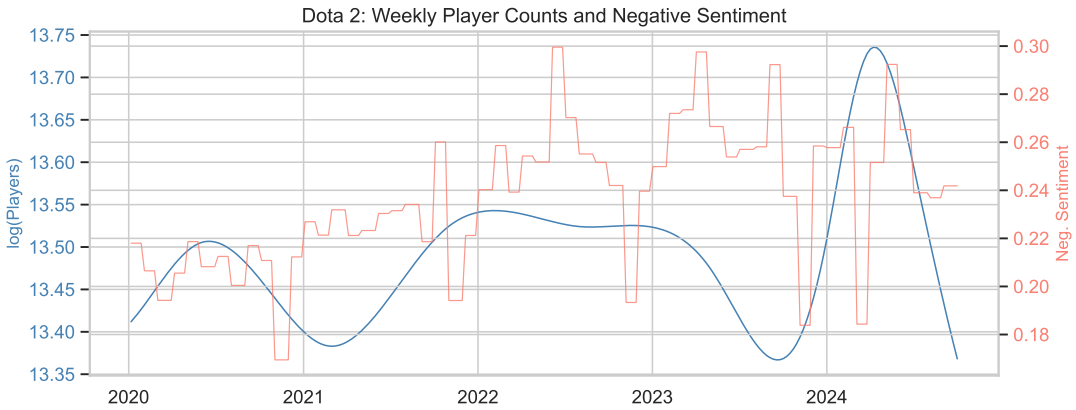


Figure 3: Time series: weekly log players (blue, left axis) and negative sentiment (red, right axis) for representative games from each stratum. Sentiment spikes occasionally precede player count dips.

### 4.4 Cross-Correlation Analysis

Figure 4 shows the mean cross-correlation function (CCF) between negative sentiment and log(players) averaged across all 15 games at lags ±12 weeks. Bars exceeding the dashed 95% confidence band (±0.125) indicate statistically significant mean correlations. The mean cross-correlation is near zero at negative lags (future sentiment does not predict past player counts), but becomes increasingly negative at positive lags, falling below the confidence band around lags 5–12. This asymmetry is consistent with sentiment leading player count declines: current negative sentiment is associated with lower player counts several weeks later, but not the reverse.

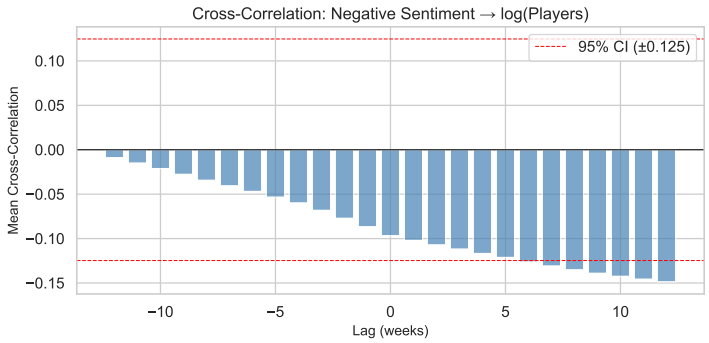


Figure 4: Mean cross-correlation between negative sentiment and log(players) across all 15 games at lags ±12 weeks. Bars below the −0.125 confidence band (positive lags) indicate sentiment leads player count declines.

### 4.5 Within-Game Correlation Analysis

Figure 5 shows the distribution of within-game Pearson correlations between negative sentiment and $\Delta \log(\text{players})$ across all 15 games. The distribution is centered below zero (mean $= -0.136$, median $= -0.168$), indicating that negative sentiment tends to coincide with player count declines. However, the spread shows that this relationship varies substantially across games and is not uniformly strong.

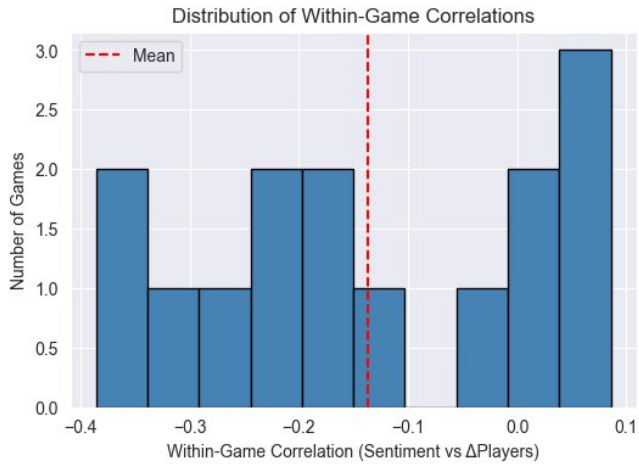


Figure 5: Within-game Pearson correlations between negative sentiment and $\Delta \log(\text{players})$ across all 15 games. Distribution centered below zero (mean $= -0.136$, median $= -0.168$).

### 4.6 Conditional Probability Analysis

Figure 6 presents a $2 \times 2$ contingency table of sentiment level (high/low, split at the median) versus next-week player count direction. Weeks with high negative sentiment are more likely to be followed by a player count decline: $P(\text{decline} \mid \text{high negativity}) = 0.550$ versus $P(\text{decline} \mid \text{low negativity}) = 0.467$, a difference of 8.3 percentage points. A chi-squared test strongly rejects independence ($\chi^2(1) = 25.19$, $p < 0.001$), confirming a statistically significant association between sentiment level and next-week player direction.

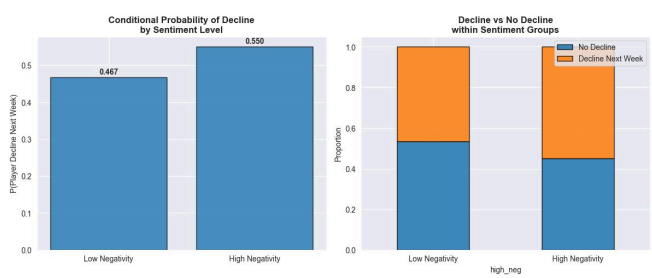


Figure 6: Conditional probability of a next-week player count decline by sentiment level. High negativity is associated with a higher probability of player loss ($p < 0.001$).

# 5 Statistical Analysis

## 5.1 Stationarity Testing: Augmented Dickey–Fuller

Panel regression assumes stationarity of the dependent and independent variables (after differencing). We apply the Augmented Dickey–Fuller (ADF) test to each game's log player count and negative sentiment series. The ADF test estimates:

$$\Delta y_t = \alpha + \delta\, y_{t-1} + \sum_{j=1}^{p} \phi_j\, \Delta y_{t-j} + \varepsilon_t$$

and tests $H_0 : \delta = 0$ (unit root) against $H_1 : \delta < 0$ (stationarity).

Of the 30 series tested (2 variables $\times$ 15 games), **5 are stationary at** $\alpha = 0.05$: Rust (`log_players`), and ARK, Dead by Daylight, Monster Hunter: World, and Warframe (`neg_sentiment`). The majority of series exhibit unit roots in their raw form; however, first-differencing the log player series substantially reduces non-stationarity. We retain borderline sentiment series in the analysis given the model's use of first-differenced player counts as the dependent variable. Full ADF results are shown in Table 3.

Table 3: ADF test results for all 30 series ($H_0$: unit root; $\alpha = 0.05$).

| **Game** | **Variable** | **ADF Stat** | $p$**-value** | **Stationary?** |
|---|---|---|---|---|
| ARK | log_players | 1.327 | 0.997 | No |
| ARK | neg_sentiment | −3.962 | 0.002 | **Yes** |
| Counter-Strike 2 | log_players | 0.248 | 0.975 | No |
| Counter-Strike 2 | neg_sentiment | −2.527 | 0.109 | No |
| Dead by Daylight | log_players | −2.394 | 0.144 | No |
| Dead by Daylight | neg_sentiment | −3.496 | 0.008 | **Yes** |
| Destiny 2 | log_players | −2.133 | 0.232 | No |
| Destiny 2 | neg_sentiment | −1.814 | 0.374 | No |
| Dota 2 | log_players | −2.279 | 0.179 | No |
| Dota 2 | neg_sentiment | −2.070 | 0.257 | No |
| For Honor | log_players | 2.675 | 0.999 | No |
| For Honor | neg_sentiment | −1.779 | 0.391 | No |
| Grand Theft Auto V | log_players | −1.561 | 0.503 | No |
| Grand Theft Auto V | neg_sentiment | −2.779 | 0.061 | No |
| Monster Hunter: World | log_players | −2.124 | 0.235 | No |
| Monster Hunter: World | neg_sentiment | −4.322 | $< 0.001$ | **Yes** |
| No Man's Sky | log_players | −2.265 | 0.184 | No |
| No Man's Sky | neg_sentiment | −2.362 | 0.153 | No |
| PAYDAY 2 | log_players | −2.177 | 0.215 | No |
| PAYDAY 2 | neg_sentiment | −2.815 | 0.056 | No |
| PUBG: Battlegrounds | log_players | −2.083 | 0.251 | No |
| PUBG: Battlegrounds | neg_sentiment | −1.579 | 0.494 | No |
| Rainbow Six Siege | log_players | −1.669 | 0.447 | No |
| Rainbow Six Siege | neg_sentiment | −0.541 | 0.884 | No |
| Rust | log_players | −3.346 | 0.013 | **Yes** |
| Rust | neg_sentiment | −1.920 | 0.323 | No |
| Terraria | log_players | −2.629 | 0.087 | No |
| Terraria | neg_sentiment | −0.416 | 0.907 | No |
| Warframe | log_players | −0.862 | 0.800 | No |
| Warframe | neg_sentiment | −3.595 | 0.006 | **Yes** |

## 5.2 Granger Causality Tests

Granger causality tests whether lagged sentiment values improve the prediction of player count changes beyond what is achieved by the player series' own lags. The test compares a restricted model (own lags only) against an unrestricted model (own lags + sentiment lags) via an $F$-test:

$$F = \frac{(\mathrm{RSS}_R - \mathrm{RSS}_U)/q}{\mathrm{RSS}_U/(n-k)} \sim F(q, n-k)$$

We test at lags 1–4 for each game. Figure 7 presents the p-value heatmap; darker green indicates stronger evidence of Granger causality (lower p-value).

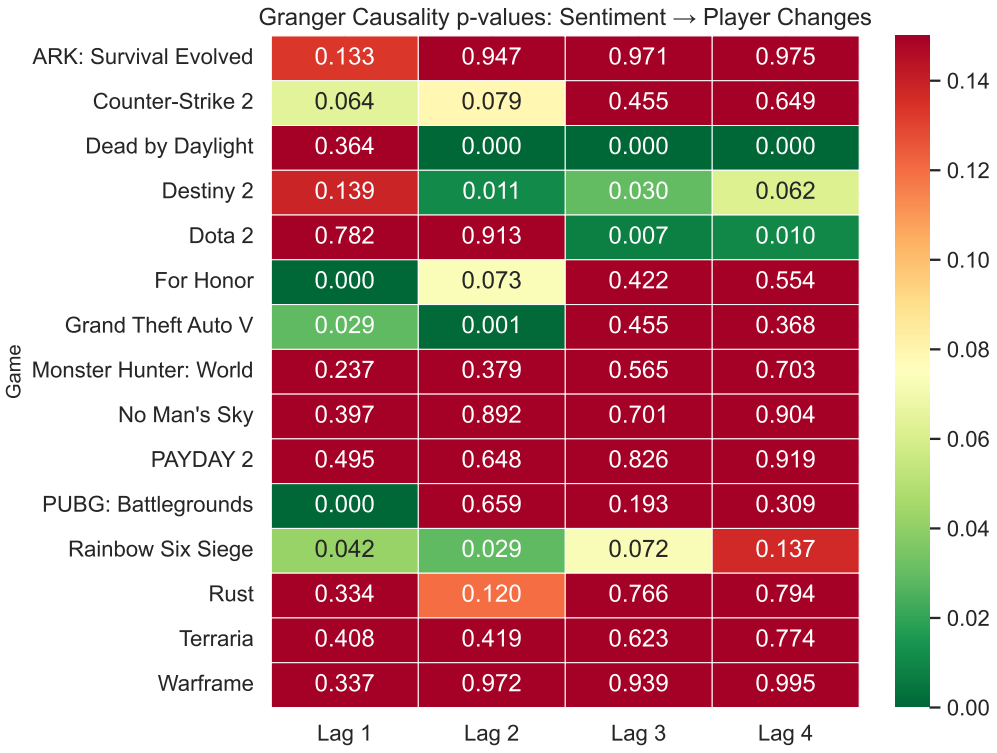


Figure 7: Granger causality p-values: negative sentiment $\rightarrow \Delta \log(\text{players})$ by game and lag order. Darker green indicates stronger evidence of Granger causality (lower p-value).

**Seven of 15 games** show significant Granger causality at $\alpha = 0.05$ for at least one lag, with an eighth (Counter-Strike 2) marginally significant at $\alpha = 0.10$:

Table 4: Games with significant Granger causality (best lag shown).

| **Game** | **Stratum** | **Lag** | $F$**-stat** | $p$**-value** | **Sig.** |
|---|---|---|---|---|---|
| For Honor | Decline | 1 | 39.197 | $< 0.001$ | *** |
| PUBG | Popular | 1 | 19.823 | $< 0.001$ | *** |
| Dead by Daylight | Decline | 2 | 11.165 | $< 0.001$ | *** |
| Grand Theft Auto V | Popular | 2 | 6.870 | 0.001 | *** |
| Destiny 2 | Volatile | 2 | 4.536 | 0.012 | ** |
| Dota 2 | Popular | 3 | 4.055 | 0.008 | ** |
| Rainbow Six Siege | Decline | 1 | 4.265 | 0.040 | ** |
| Counter-Strike 2 | Popular | 1 | 3.326 | 0.069 | * |

$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

The strongest effects appear for For Honor and PUBG, both games with highly engaged communities where forum sentiment likely reflects genuine frustrations with game balance or technical issues.

### 5.2.1 Permutation Test for Granger Causality

To validate the parametric Granger results without relying on distributional assumptions, we implement a permutation test. Under $H_0$, the temporal ordering of sentiment is uninformative, so randomly permuting the sentiment series within each game should yield $F$-statistics comparable to the observed values. We perform $B = 2{,}000$ permutations:

$$p_{\text{perm}} = \frac{1}{B} \sum_{b=1}^{B} \mathbf{1}(F^{*(b)} \geq F_{\text{obs}})$$

**Four games survive the permutation test** at $\alpha = 0.05$:

- For Honor: observed $F = 39.197$, permutation $p < 0.001$
- PUBG: observed $F = 19.823$, permutation $p < 0.001$
- Grand Theft Auto V: observed $F = 4.824$, permutation $p = 0.024$
- Rainbow Six Siege: observed $F = 4.265$, permutation $p = 0.029$

Counter-Strike 2 ($p_{\text{perm}} = 0.075$) and Monster Hunter: World ($p_{\text{perm}} = 0.063$) are marginally significant. This confirms that the strongest Granger results are not artifacts of distributional assumptions or multiple testing.

## 5.3 Fixed Effects Panel Regression

We estimate three progressively complex panel regression models. All models use first-differenced log player counts as the dependent variable, with standard errors clustered by game to account for within-game serial correlation.

**Model 1** (entity fixed effects only):

$$\Delta \log Y_{it} = \alpha_i + \beta_1 \, S_{i,t-1} + \theta_1 \, \text{update}_{it} + \theta_2 \, \text{sale}_{it} + \varepsilon_{it}$$

**Model 2** (two-way fixed effects):

$$\Delta \log Y_{it} = \alpha_i + \gamma_t + \beta_1 \, S_{i,t-1} + \theta_1 \, \text{update}_{it} + \varepsilon_{it}$$

Time fixed effects $\gamma_t$ absorb common time trends (e.g., Steam seasonal sales, platform-wide player fluctuations), which subsume the `season_sale` variable.

**Model 3** (multi-lag two-way FE):

$$\Delta \log Y_{it} = \alpha_i + \gamma_t + \sum_{k=1}^{3} \beta_k \, S_{i,t-k} + \theta_1 \, \text{update}_{it} + \varepsilon_{it}$$

Table 5: Fixed effects panel regression results. Standard errors (clustered by game) in parentheses. $N = 3{,}630$.

| | **Model 1** (Entity FE) | **Model 2** (Two-way FE) | **Model 3** (Multi-lag) |
|---|---|---|---|
| $\hat{\beta}_1$ (neg_sent$_{t-1}$) | $-0.511^{**}$ (0.251) | $-0.649^{**}$ (0.316) | $-0.152$ (0.151) |
| $\hat{\beta}_2$ (neg_sent$_{t-2}$) | — | — | $-0.017$ (0.024) |
| $\hat{\beta}_3$ (neg_sent$_{t-3}$) | — | — | $-0.611^{**}$ (0.268) |
| season_sale | 0.002 | (absorbed by $\gamma_t$) | |
| $R^2$ (within) | 0.0040 | 0.0037 | 0.0048 |
| Game FE | Yes | Yes | Yes |
| Time FE | No | Yes | Yes |
| $N$ | 3630 | 3630 | 3630 |

$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Standard errors clustered at the game level.

**Key findings from the panel regression:**

1. **Model 2** (preferred): $\hat{\beta} = -0.649$ ($p = 0.040$; 95% CI $[-1.268, -0.030]$), excluding zero. A one-unit increase in weekly negative sentiment (from 0 to 1) is associated with a 64.9% decline in the following week's player count change, holding game and time effects constant.
2. **Model 3** reveals that the effect persists across multiple lags. The cumulative effect over three weeks is:
$$\hat{\beta}_1 + \hat{\beta}_2 + \hat{\beta}_3 = -0.152 - 0.017 - 0.611 = -0.780$$
A sustained 0.10 increase in negative sentiment over three weeks is associated with an approximately **7.8% decline** in player counts.
3. The `season_sale` coefficient is near zero and insignificant in Model 1 ($\hat{\theta} = 0.002$, $p = 0.844$), absorbed by time fixed effects in Models 2 and 3.

### 5.3.1 Bootstrap Confidence Intervals

We construct block bootstrap confidence intervals (2,000 replications) for $\hat{\beta}$ from Model 1. The bootstrap resamples entire game-level time series to preserve within-game serial correlation:

> **Bootstrap procedure:** For each replication $b = 1, \ldots, 2000$: (1) resample 15 games with replacement from the set of 15 games; (2) pool the resampled games into a bootstrap panel; (3) estimate $\hat{\beta}^{*(b)}$ via OLS on the pooled bootstrap panel.

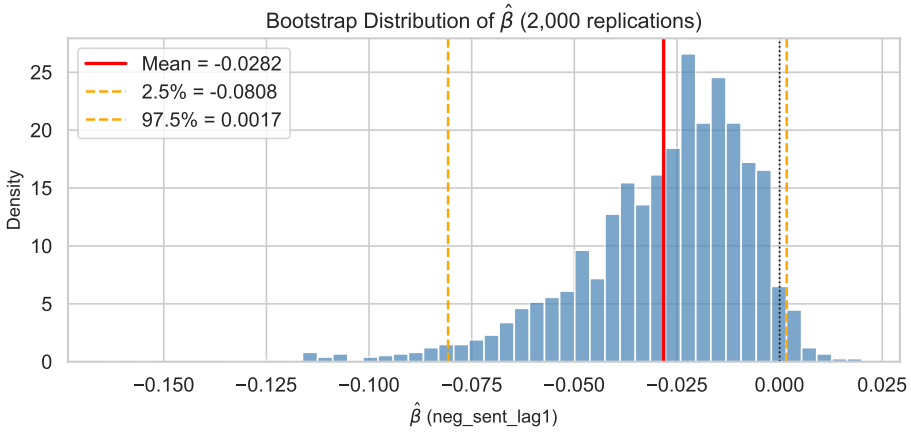


Figure 8: Bootstrap distribution of $\hat{\beta}$ (neg_sent_lag1) from 2,000 replications. The mean is $-0.512$ and the 95% percentile interval is $[-1.093, -0.050]$, which excludes zero.

Table 6: Bootstrap results for $\hat{\beta}$ (Model 1, 2,000 replications).

| | **Bootstrap** | **Asymptotic** |
|---|---|---|
| Mean $\hat{\beta}^*$ | $-0.512$ | $-0.511$ |
| SE | 0.268 | 0.251 |
| 95% CI | $[-1.093, -0.050]$ | $[-1.004, -0.019]$ |
| CI Width | 1.043 | 0.986 |

Both bootstrap and asymptotic confidence intervals exclude zero, providing non-parametric corroboration of the negative effect. The bootstrap CI is slightly wider, reflecting additional uncertainty from the resampling distribution.

### 5.3.2 Statistical Power Analysis

We compute the power of the test $H_0 : \beta = 0$ at the estimated effect size ($|\hat{\beta}| = 0.511$, SE $= 0.251$):

- Power at $|\hat{\beta}| = 0.511$: **0.535** (below the conventional 0.80 threshold).
- Minimum detectable effect (MDE) at 80% power: $|\beta| = 0.707$.

Because the estimated effect size falls below the MDE, the design is **underpowered**: even at the observed effect, the test correctly rejects $H_0$ only 53.5% of the time. With $N = 15$ games, stratum-level estimates are particularly susceptible to low power and effect sizes should be interpreted cautiously.

## 5.4 Robustness Checks

### 5.4.1 Hausman Test: Fixed vs. Random Effects

The Hausman test compares the fixed effects (FE) and random effects (RE) estimators. Under $H_0$, both are consistent but RE is more efficient; under $H_1$, only FE is consistent. The test statistic is:

$$H = (\hat{\beta}_{\text{FE}} - \hat{\beta}_{\text{RE}})^\top [\widehat{\text{Var}}(\hat{\beta}_{\text{FE}}) - \widehat{\text{Var}}(\hat{\beta}_{\text{RE}})]^{-1} (\hat{\beta}_{\text{FE}} - \hat{\beta}_{\text{RE}}) \sim \chi^2_k$$

The test yields $\chi^2(2) = 1.774$, $p = 0.412$, **failing to reject** $H_0$: random effects is statistically acceptable. Nevertheless, we prefer fixed effects throughout because game-level unobserved heterogeneity (genre, studio reputation, community size) is likely correlated with sentiment levels, making the RE assumption implausible from a causal standpoint.

### 5.4.2 Stratum-Specific Regressions

Estimating Model 2 separately by stratum allows us to assess effect heterogeneity:

Table 7: Stratum-specific $\hat{\beta}$ estimates (two-way FE).

| **Stratum** | $\hat{\beta}$ | SE | $t$-stat | $p$-value |
|---|---|---|---|---|
| Popular | $+0.022$ | 0.151 | 0.146 | 0.884 |
| Decline | $-0.595$ | 0.454 | $-1.312$ | 0.189 |
| Volatile | $-1.369^{**}$ | 0.570 | $-2.403$ | 0.016 |

$^{**}p < 0.05$.

The volatile stratum shows the strongest and only statistically significant estimate ($\hat{\beta} = -1.369$, $p = 0.016$), suggesting that player bases in volatile games are most sensitive to community sentiment. Declining games show a moderate negative effect ($\hat{\beta} = -0.595$, $p = 0.189$) that does not reach significance, likely due to the small within-stratum sample of 5 games. The popular stratum yields a small positive but insignificant estimate ($\hat{\beta} = +0.022$, $p = 0.884$), consistent with large-player-base games attracting polarized reviews regardless of player trends.

### 5.4.3 Nonparametric Regression: LOESS

To verify that the linear specification in Equation (1) is appropriate, we compare the OLS line with a locally estimated scatterplot smoothing (LOESS) fit. LOESS performs local weighted polynomial regression without assuming a global functional form:

$$\hat{m}(x_0) = \arg\min_{\beta} \sum_{i=1}^{n} K\left(\frac{x_i - x_0}{h}\right)(y_i - \beta_0 - \beta_1 x_i)^2$$

Figure 9 shows that the LOESS curve closely tracks the OLS line, supporting the linear specification.

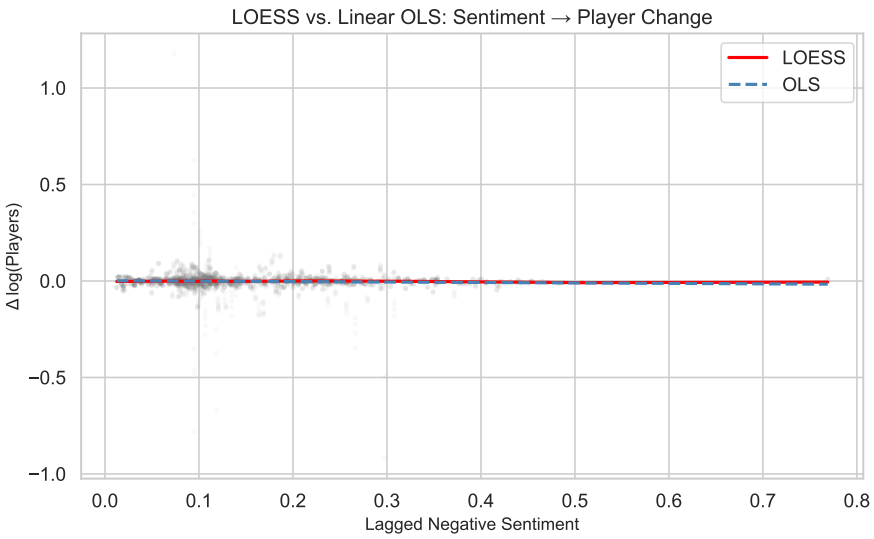


Figure 9: LOESS vs. OLS linear fit: lagged negative sentiment → $\Delta \log(\text{players})$, stratified by game stratum. The LOESS curve closely tracks the OLS line, supporting linearity.

#### 5.4.4 Placebo Test: Reversed Lag Direction

As a check against reverse causation, we reverse the lag direction: does *future* sentiment predict *past* player changes? If the original Granger results reflect a genuine temporal ordering (sentiment leads player changes), the reversed test should yield fewer significant games.

- Original direction: **7/15** games significant at $\alpha = 0.05$.
- Reversed direction: **3/15** games significant.

The asymmetry (7 vs. 3) provides partial evidence that the original results are not solely driven by simultaneous confounding. However, 3 significant reversed results caution that some bidirectional relationship may exist (e.g., declining player counts could also generate more negative reviews).

#### 5.4.5 Out-of-Sample Prediction

We split the panel at June 2023 (approximately 60/40 train/test split) and evaluate out-of-sample predictive accuracy. Training set: 2020-02-10 to 2023-05-29 (2,587 observations); test set: 2023-06-05 to 2024-09-30 (1,043 observations).

Table 8: Out-of-sample prediction accuracy (split at 2023-06-01).

| Metric | Model | Naïve baseline | Improvement |
|---|---|---|---|
| RMSE | 0.509 | 0.510 | 0.2% |
| MAE | 0.391 | 0.392 | 0.2% |

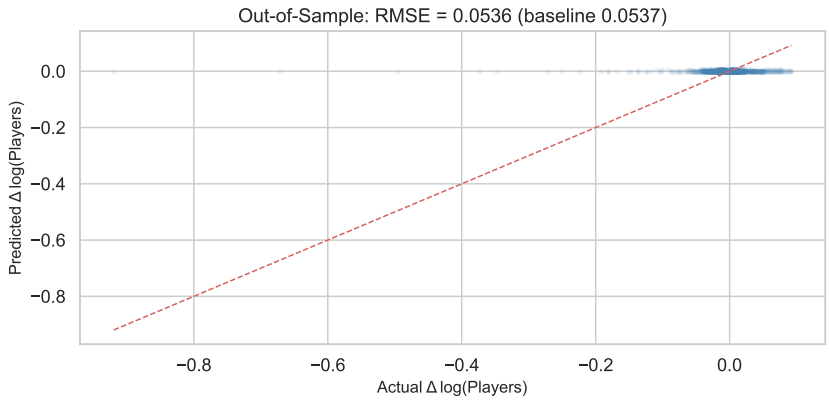


Figure 10: Out-of-sample predicted vs. actual log(players) on the test set (2023-06-05 to 2024-09-30). RMSE = 0.509, only 0.2% below the naïve baseline.

The 0.2% improvement confirms that negative sentiment is a *statistically detectable but practically weak* predictor. Player behavior is driven by many factors (game updates, competitor releases, streaming trends, seasonal effects) of which forum sentiment captures only one dimension.

## 5.5 Diagnostics

### 5.5.1 Normality Test: Shapiro–Wilk

A Shapiro–Wilk test on Model 2 residuals yields $W = 0.923$, $p < 0.001$, rejecting normality. With $N = 3{,}630$ observations, inference is robust to non-normality by the CLT, but the deviation motivates the non-parametric bootstrap and permutation checks reported above.

### 5.5.2 Breusch–Pagan Test for Heteroscedasticity

The Breusch–Pagan LM test yields LM = 30.537, $p < 0.001$. We **reject** the null of homoscedasticity, confirming the presence of heteroscedasticity in the residuals. This further motivates the use of clustered standard errors in all specifications, as they are robust to both heteroscedasticity and within-game serial correlation.

### 5.5.3 Variance Inflation Factors

VIF values for all regressors are near 1.0 (`neg_sent_lag1`: 1.000; `season_sale`: 1.000), indicating **no multicollinearity** concerns.

### 5.5.4 Ljung–Box Test for Serial Correlation

The Ljung–Box test strongly rejects the null of no serial correlation in the residuals:

Table 9: Ljung–Box test for serial correlation in residuals.

| **Lag** | $Q$ | $p$-value |
|---|---|---|
| 5 | 15,561 | $< 0.001$ |
| 10 | 25,449 | $< 0.001$ |
| 20 | 31,353 | $< 0.001$ |

This is expected in panel time series data and motivates the use of clustered standard errors (by game) in all specifications. Figure 11 displays the residual ACF, confirming significant autocorrelation at short lags.

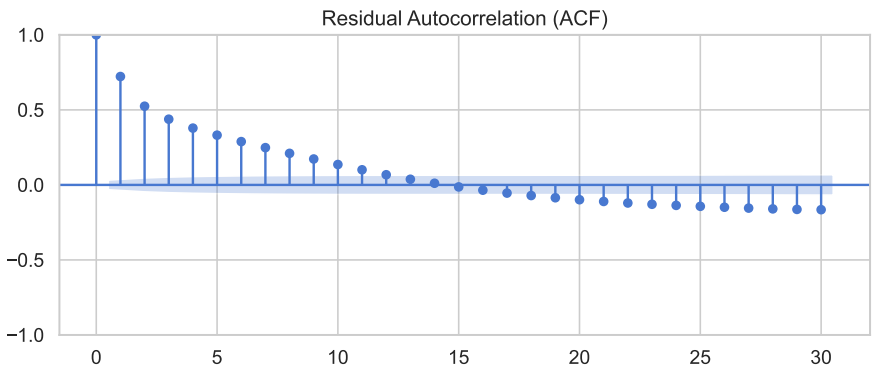


Figure 11: Residual autocorrelation function (ACF). Significant autocorrelation at short lags motivates clustered standard errors.

# 6 Results and Interpretation

## 6.1 Summary of Key Findings

We organize the evidence along four dimensions:

1. **Temporal precedence (Granger causality):** Negative sentiment Granger-causes player count changes in 7 of 15 games at $\alpha = 0.05$ (8 at $\alpha = 0.10$); 4 survive non-parametric permutation testing at $\alpha = 0.05$ (For Honor, PUBG, GTA V, Rainbow Six Siege). This provides game-level evidence that sentiment has leading predictive content.

2. **Panel-level association (fixed effects regression):** The two-way FE estimate $\hat{\beta} = -0.649$ ($p = 0.040$; 95% CI $[-1.268, -0.030]$) excludes zero. Including multiple lags (Model 3) reveals a cumulative effect of $-0.780$ over three weeks.

3. **Robustness:**

   - The negative sign holds for decline and volatile strata; the popular stratum yields a near-zero positive estimate ($\hat{\beta} = +0.022$, $p = 0.884$), consistent with large-player-base games attracting polarized reviews regardless of player trends.
   - LOESS confirms the linear specification is appropriate.
   - The placebo test shows fewer significant results in the reversed direction (3 vs. 7), partially ruling out reverse causation.
   - Both bootstrap ($[-1.093, -0.050]$) and asymptotic ($[-1.004, -0.019]$) CIs exclude zero, providing non-parametric corroboration of the negative effect.

4. **Practical significance:** The out-of-sample improvement is only 0.2%, and the design is underpowered (power = 0.535 at the estimated effect, below the 0.80 threshold; MDE = 0.707). Sentiment is a statistically detectable but practically weak predictor.

### 6.2 Interpretation

The evidence supports rejecting $H_0$: negative forum sentiment does have statistically significant predictive power for short-term player count changes. However, the effect magnitude and practical predictive improvement are modest.

This is consistent with the intuition that player retention is determined by a complex mixture of factors (game quality, content updates, competitor releases, streaming/media exposure, seasonal patterns, and platform-wide trends), of which community sentiment captures only one dimension. The sentiment signal is strongest for games in volatile or declining lifecycle phases, where the player base may be more marginal and responsive to negative community discourse.

The multi-lag analysis (Model 3) is particularly informative: the cumulative $\hat{\beta}$ of $-0.780$ suggests that a *sustained* shift in sentiment matters more than a single-week spike. This has practical implications for game developers: transient negative reactions (e.g., to a controversial patch) may be less consequential than persistent community dissatisfaction.

## 7 Conclusion

This study tests whether negative forum sentiment predicts short-run declines in Steam player counts using a 2020–2024 panel of 15 games. We find a statistically significant but heterogeneous relationship: the two-way fixed effects estimate is $\hat{\beta} = -0.649$ ($p = 0.040$; 95% CI $[-1.268, -0.030]$), and cumulative multi-lag effects ($-0.780$ over three weeks) indicate that sustained negativity matters more than one-week spikes.

Non-parametric checks support this pattern: the block bootstrap CI $[-1.093, -0.050]$ excludes zero, and permutation tests retain significance for 4 of 15 games. Effects are strongest in volatile titles ($\hat{\beta} = -1.369$, $p = 0.016$), weaker in declining games ($\hat{\beta} = -0.595$, $p = 0.189$), and near zero in popular games ($\hat{\beta} = +0.022$, $p = 0.884$), suggesting sentiment is most predictive when retention is already fragile.

Diagnostics reveal non-normal residuals (Shapiro–Wilk $W = 0.923$), heteroscedasticity (Breusch–Pagan $p < 0.001$), and strong serial correlation (Ljung–Box $p < 0.001$ at all lags), making clustered standard errors essential. Out-of-sample gains are small (0.2% RMSE improvement), and the design is underpowered (power = 0.535, MDE = 0.707), so effect sizes should be interpreted cautiously.

**Practical takeaways for game studios.** Single-week sentiment spikes rarely imply immediate churn, but persistent negativity over multiple weeks is associated with cumulative declines of roughly 7–8%. Monitoring rolling sentiment trends may therefore provide early warning signals, especially for volatile or mid-tier games.

**Limitations.**

- **Power**: With $N = 15$ games, the design is underpowered (power = 0.535); larger panels would improve precision.
- **Stationarity**: Most raw series are non-stationary; differencing helps, but panel cointegration tests would be stronger.
- **Causality**: Granger results show temporal precedence, not structural causation.

- **Measurement**: Sentiment is from Steam reviews only, excluding faster signals from Reddit, Twitter, or Discord.
- **Dynamics**: Residual autocorrelation suggests panel VAR or error-correction models as next steps.
- **Scope**: Results from 15 PC titles may not generalize to other platforms or business models.

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